Roll No. | | | | | | | |

End Semester Examination 2025

Name of the Course: ***M. Tech (Electric Vehicle Technology)*** Semester: 1st
Name of the Paper – Electrical Machines and Drives
Paper Code: MEV 104

Time: 3Hours Maximum Marks: 100

Note: -

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Explain the basic concept of an electric drive. Discuss the advantages of electrical drives over other types of drives.	
(b)	With the help of a neat block diagram, explain the basic structure of an electric drive system. Describe the function of each block in detail.	CO1
(c)	Discuss at least five major advantages of electrical drives that make them widely used in modern industries.	
Q2		
(a)	Derive the fundamental torque equation of an electrical drive and explain the speed–torque conventions used for motoring and braking operations. Illustrate your answer with a multi-quadrant speed–torque diagram.	
(b)	What are the components of load torque acting on an electrical drive? Discuss the nature and classification of load torques with suitable examples from industrial applications.	CO2
(c)	A drive has the following parameters: J=10 kg-m2, T=100-0.1*N*, N-m, where *N* is the speed in rpm. Intially the drive is operating in steady state. Now it is to be reversed. For this motor characteristics is changed to -100-0.1*N*, N-m. Calculate time of reversal.	
Q3		CO3
(a)	With the help of a neat circuit diagram and waveform, explain the operation of a single-phase fully-controlled rectifier-fed DC separately excited motor drive. Derive the expression for average armature voltage.	
(b)	Describe the chopper control of a separately excited DC motor with a neat circuit diagram. Discuss the speed–torque characteristics under different duty cycles.	
(c)	A 230V, 960 rpm, and 200 A, separately excited dc motor has an armature resistance of 0.02 Ω. The motor is fed from a chopper which provides both motoring and braking operation. The source has a voltage of 230 V. Assuming continuous conduction Calculate (i) duty ratio of chopper for motoring operation at rated torque and 350 rpm. (ii) duty ratio of chopper for braking operation at rated torque and 350 rpm.	
Q4		CO4
(a)	Explain the principle of variable frequency control from a current source. Discuss how varying the frequency and current affects the torque–speed characteristics of an induction motor.	

(b)	Derive the relationship between supply frequency, motor speed, and torque for a current-source inverter-fed induction motor. Discuss how the control strategy maintains constant torque operation.	
(c)	Starting from the equivalent circuit, derive the torque–slip characteristic equation of an induction motor. Sketch the typical torque–slip curve and explain the regions of motoring, generating, and braking.	
Q5		**CO5, CO6**
(a)	Explain the operation of a variable-speed constant-frequency (VSCF) system using a synchronous generator.	
(b)	A 2.8 kW, 400 V, 50 Hz, 4 pole, 1370 rpm, delta connected squirrel cage induction motor has the following parameters referred to the stator: Rs= 2 Ω. Rr'=5 Ω Xs=Xr'= 5 Ω, Xm=80 Ω. Motor speed is control by stator voltage control. When driving a fan load it runs at a rated speed at rated voltage. Calculate motor terminal voltage current and torque at 1200 rpm.	
(c)	A 2.8 kW, 400 V, 50 Hz , 4 pole, 1370 rpm, delta connected squirrel cage induction motor has the following parameters referred to the stator: Rs= 2 Ω. Rr'=5 Ω Xs=Xr'= 5 Ω, Xm=80 Ω. Motor speed is control by stator voltage control .Calculate motor speed, current and torque for the terminal voltage of 300 V	